Review 2 Commitment

1. Project Information

Project code	FA26SE032
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
Review 2 date	29/10/2026
Reviewer 1	longt5
Reviewer 2	ThinhDP2
Version	2.0
Date	07/11/2026

2. Final Registration Form

Final version	Version 2.1 (Post-Review 2 Realigned Enterprise In-House Affiliate + Web & Mobile Scope)
Date confirmed	06/11/2026
Record of Changes (BM03)	Form BM03 bundled in SWP490_FA26SE032_R2_Registration.pdf (6 realignments).

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-01	Multi-Role Auth, Registration/Reset Email OTP & 5-Role RBAC (Web & Mobile)	All 5 Roles	Argon2id (>=64MB) hashes passwords; 6-digit Email OTP verifies new registrations & password resets (>=5 wrong OTPs invalidate code); Login alerts via email; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden.	WF-01	Nguyễn Thành Thắng
F-02	Internal Brand Staff & In-House Store Provisioning	System Administrator	Only SYSTEM_ADMIN creates SHOP_MANAGER (bound to storeId & GHN warehouse) and SYSTEM_MANAGER; public /auth/register with SHOP_MANAGER coerces to CUSTOMER.	WF-01	Nguyễn Thành Thắng
F-03	KOC Upgrade eKYC Maker-Checker & Social Channels	Customer, Collaborator, System Manager	Customer submits 12-digit CCCD, bank account & TikTok/FB/YouTube URL; SYSTEM_MANAGER approval sets KycStatus=VERIFIED, role=COLLABORATOR, starting tier=Bronze (10%).	WF-01	Nguyễn Thành Thắng
F-04	Immutable Security & Financial Audit Trail	System Administrator, System Manager	100% of staff creations, KYC decisions, MAP updates, VIP deal approvals, return resolutions & payouts append an AuditLog entry; API blocks edit/delete.	WF-01	Nguyễn Thành Thắng
F-05	Category-Locked 6-Block Product & Compliance Moderation	Shop Manager, System Manager	Product category locks to store.category (mismatch returns 400); duplicate EAN-13 barcode fails; new SKU stays DRAFT until a System Manager approves (APPROVED).	WF-02	Nguyễn Thành Thắng
F-06	MAP Floor-Price Guard & Price Promotion Approval	Shop Manager, System Manager, System Administrator	System Manager sets SKU floorPrice; single/batch promotion with proposedPrice < floorPrice returns 400; discount applies only when status=ACTIVE within budget.	WF-02	Nguyễn Thành Thắng
F-07	4-Axis AI Partner Matchmaking & Two-Way Cooperation	Shop Manager, Collaborator	AI scores verified KOCs (0–100 across Category, CR%, Tier/Social, Price Fit) in <=500ms; bilateral partnership invite/request transitions PENDING -> APPROVED.	WF-02	Nguyễn Thành Thắng

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F-08	Socket.IO Deal Chat & Realtime Media/File Sharing	Shop Manager, Collaborator	Real-time bi-directional chat allowing Shop Managers and Collaborators to negotiate terms and directly exchange product photos, review guidelines, and HD marketing assets without a static media hub.	WF-02	Nguyễn Thành Thắng
F-09	Socket.IO Realtime VIP Deal Negotiation (10%–16%)	Shop Manager, Collaborator	Counterparties send/respond to rich VIP Deal cards (targetRevenue, commitmentDays) over /chat WebSocket without reload; 10%–16% deals activate immediately on accept.	WF-03	Phan Xuân Thịnh
F-10	VIP Budget Exception Escalation Gate (17%–25%)	Shop Manager, Collaborator, System Administrator	Proposals >25% fail with 400; accepted 17%–25% deals transition to PENDING_ADMIN_APPROVAL and activate only after SYSTEM_ADMIN approves at /admin/vip-exceptions.	WF-03	Phan Xuân Thịnh
F-11	Sample Product Sponsorship & 14-Day Video Review SLA	Collaborator, Shop Manager, System Manager	Verified KOC requests 0 VND sample; GHN receipt (RECEIVED) starts 14-day SLA timer; unsubmitted video past deadline marks OVERDUE and blocks new sample requests.	WF-03	Phan Xuân Thịnh
F-12	Value-Added Livestream Room & Live Deal Pinning	Collaborator, Shop Manager, Customer	Creator transitions SCHEDULED -> LIVE -> ENDED and pins store SKUs (+2% live bonus); stock=0 disables buy CTA in live tray; ENDED locks pinning and shows summary.	WF-03	Phan Xuân Thịnh
F-13	Smart Link, Dynamic QR & KOC Attribution Coupon	Collaborator, Shop Manager, Customer	Verified KOC creates /r/:shortCode link, QR & goods coupon ([KOC][5|10], shipping vouchers excluded); click sets 30d AttributionSession rate-limited at 100 req/min/IP.	WF-04	Nguyễn Phú Quý
F-14	Authenticated Zero-Loss Checkout & Self-Referral Guard	Customer, Collaborator	Checkout requires login while preserving 100% cart items and settles via PayOS VietQR/Wallet/COD; if buyer matches KOC (isSelfReferral), commission is forced to 0%.	WF-04	Nguyễn Phú Quý
F-15	Logistics Status Ingestion & 14-Day Commission Hold Sync	Shop Manager, Customer	Webhook syncs GHN parcel status (SHIPPING -> DELIVERED); delivery event atomically starts 14-day return window and holds commission in PENDING_14D_HOLD.	WF-04	Nguyễn Phú Quý
F-16	Dual-Path Return Resolution: Exchange vs Refund Clawback	Customer, Shop Manager, System Manager	Within 14-day return window: carrier/factory damage 1-for-1 EXCHANGE preserves 100% KOC commission; customer REFUND triggers automated Clawback of commission to 0 VND and deducts GMV from active Deal KPI.	WF-04	Nguyễn Phú Quý
F-17	Two-Stage Non-Stacking Commission, 14-Day Hold & KPI Deal Escrow	Shop Manager, Collaborator, System Manager	Enforces non-stacking two-stage settlement: Stage 1 matures 10% Base + Tier% to availableBalance after 14 days clean delivery; Stage 2 holds KPI differential max(0, Deal% - (10%+Tier%)) in KPI Escrow until 100% KPI achievement.	WF-05	Nguyễn Đình Tuấn
F-18	Monthly Milestone Settlement (FR-09) & Tier Promotion	Shop Manager, Collaborator	Closed-month clean GMV scan excludes refunded/self-referral orders, pays 200k–4M VND milestone bonus idempotently (0 duplicates), and promotes Bronze->Silver->Gold->Diamond.	WF-05	Nguyễn Đình Tuấn
F-19	Row-Locked Collaborator Withdrawal, 10% PIT & Napas 24/7 Payout	Collaborator, System Manager, System Administrator	SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200,000 VND) locks wallet row, withholds flat 10% PIT (FinancePosting), and disburses net via Napas 24/7 Sandbox.	WF-05	Nguyễn Đình Tuấn

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-01	API & Redirect Latency	REST & Smart Link p95 Response Time	<= 200 ms for catalog queries and /r/:shortCode redirection	Autocannon benchmark (500 concurrent requests) & NestJS timing interceptor	Nguyễn Thành Thắng
NFR-02	Authentication & RBAC Security	Password Hash Cost & Role Isolation	Argon2id (>= 64 MB memory, 3 iterations); 100% cross-role calls blocked (403)	Jest security suites (auth-otp.spec.ts, rbac guards) across Web & Mobile endpoints	Nguyễn Thành Thắng

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
NFR-03	Web & Mobile Client Responsiveness	Web Initial Bundle / FCP & Mobile Bootstrap	Web gzipped entry <= 500 KB, FCP <= 1.5 s; Android /mobile/bootstrap <= 300 ms	Vite build bundle audit, Lighthouse CI, and Mobile contract/startup test suite	Phan Xuân Thịnh
NFR-04	Transactional & Inventory Integrity	Atomic Checkout & Return Rollback Rate	100% atomic rollback on stock exhaustion or invalid return state transition	Jest integration specs (orders.service.spec.ts, return-pickup.postgres.integration.spec.ts)	Nguyễn Phú Quý
NFR-05	Rate Limiting & Cache Resilience	Click Throttle & Redis Outage Failover	100 req/min/IP limit; 0 unhandled 500 errors during simulated Redis disconnect	CacheService LRU memory fallback unit tests & click-flood load script	Nguyễn Đình Tuấn
NFR-06	Financial Concurrency & Auditability	Double-Spend Prevention & Ledger Coverage	0 negative wallet balances under 20 parallel payouts; 100% mutations in AuditLog	Parallel SELECT ... FOR UPDATE payout concurrency test & FinancePosting assertions	Nguyễn Đình Tuấn

5. Out of Scope

Item	Reason
Open External Seller Marketplace & Public Shop Registration	SCANMS governs external collaborators for in-house enterprise brand divisions; external third-party merchants cannot open public stores.
Generic Retail Voucher Stacking & Standalone Marketing Portals	Platform/freeship voucher stacking and standalone promotional portals are excluded to prevent margin cannibalization; only MAP-guarded ProductPricePromotion and KOC-bound attribution coupons are supported.
Factory Manufacturing, Supplier Procurement (ERP/AP) & Robotics WMS	Upstream OEM manufacturing, purchase orders, and warehouse hardware belong to corporate ERP; SCANMS manages sellable stock quotas and SKU floor prices.
Proprietary Courier Vehicles & Licensed Core Banking Rails	Physical parcel delivery and fiat banking rails are delegated to GHN Sandbox API, PayOS (order collection), and Napas 24/7 Sandbox API (collaborator payouts).

6. Workflow Commitments

6.1 Workflow WF-01

Workflow name	WF-01: Internal Brand Staff Provisioning, KOC eKYC Onboarding & Social Verification
Owner (one member)	Nguyễn Thành Thắng
Supporting members	Phan Xuân Thịnh, Nguyễn Đình Tuấn
Actors	System Administrator, Customer, System Manager, Collaborator
Trigger	System Admin appoints an internal Brand Manager (/admin/personnel), or a Customer registers, verifies Email OTP, and submits a KOC upgrade eKYC dossier (/customer/upgrade).
Business outcome	In-house Shop Manager bound to assigned brand store; verified Customer upgraded to Collaborator at Bronze (10%) tier with verified social channel and immutable AuditLog.
Main entity state path	User: isActive=true (SHOP_MANAGER/COLLABORATOR); KycStatus: UNVERIFIED -> VERIFIED / REJECTED; Tier: Bronze (10%)
Activity diagram (Report 4 section)	Report 4 Section 3.2.1 (Activity Diagram AD-WF01)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-01	Multi-Role Auth, Registration/Reset Email OTP & 5-Role RBAC (Web & Mobile)	Argon2id (>=64MB) hashes passwords; 6-digit Email OTP verifies new registrations & password resets (>=5 wrong OTPs invalidate code); Login alerts via email; unauthorized Web routes or /api/mobile/bootstrap calls return 403 Forbidden.	TC-F01-01..03
F-02	Internal Brand Staff & In-House Store Provisioning	Only SYSTEM_ADMIN creates SHOP_MANAGER (bound to storeId & GHN warehouse) and SYSTEM_MANAGER; public /auth/register with SHOP_MANAGER coerces to CUSTOMER.	TC-F02-01..03
F-03	KOC Upgrade eKYC Maker-Checker & Social Channels	Customer submits 12-digit CCCD, bank account & TikTok/FB/YouTube URL; SYSTEM_MANAGER approval sets KycStatus=VERIFIED, role=COLLABORATOR, starting tier=Bronze (10%).	TC-F03-01..03
F-04	Immutable Security & Financial Audit Trail	100% of staff creations, KYC decisions, MAP updates, VIP deal approvals, return resolutions & payouts append an AuditLog entry; API blocks edit/delete.	TC-F04-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	**Main**	Seeded System Admin, System Manager, and new Customer account; Sora Skincare store configured.	1. Admin creates internal SHOP_MANAGER bound to Sora Skincare at /admin/personnel. 2. Customer registers, verifies Email OTP, and submits CCCD, MB Bank & TikTok channel at /customer/upgrade. 3. System Manager approves eKYC on Web/Mobile.	Staff account active on Sora Skincare; Customer upgraded to COLLABORATOR with KycStatus=VERIFIED, initial tier Bronze (10%), and AuditLog recorded.	Nguyễn Thành Thắng
DS-01b	**Exception**	Unauthenticated client sends role=SHOP_MANAGER to /auth/register; KOC submits blurred CCCD or >5 wrong registration OTPs.	1. Call POST /auth/register with role=SHOP_MANAGER. 2. Enter wrong registration OTP 5 times. 3. System Manager rejects mismatched KYC bank/CCCD dossier with mandatory reason.	Public signup coerced to CUSTOMER (privilege escalation blocked); registration OTP invalidated after 5 failures; KycStatus=REJECTED with feedback and resubmission enabled.	Nguyễn Thành Thắng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	14	Yes (>= 90%)	Nguyễn Thành Thắng

Test type	Count	Automated in CI	Responsible member(s)
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Thành Thắng
State transition tests (>= 1 per transition)	2	Yes (100%)	Nguyễn Thành Thắng
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Thành Thắng
Total for this workflow	**18**	**Yes (>= 90%)**	**Nguyễn Thành Thắng**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.2 Workflow WF-02

Workflow name	WF-02: Category-Locked Product Compliance, MAP Floor-Price Promotion, AI Matchmaking & Socket.IO Deal Negotiation
Owner (one member)	Phan Xuân Thịnh
Supporting members	Nguyễn Thành Thắng, Nguyễn Phú Quý
Actors	Shop Manager, System Manager, Collaborator
Trigger	Shop Manager manages 2-tier product variants with Floor Price guard (floorPrice), runs 4-Axis AI Partner Matchmaking, and initiates bi-directional Exclusive Deal negotiation via real-time Socket.IO chat (with direct media/file sharing).
Business outcome	Category-locked catalog and variants approved, MAP floorPrice strictly enforced, top KOC matched, and Exclusive Deal agreed (<=16% autonomous Shop Manager approval; >16% escalated to System Admin).
Main entity state path	ProductModerationStatus: DRAFT -> APPROVED / REJECTED; PricePromotionStatus: PENDING -> APPROVED -> ACTIVE; StoreCollaboratorStatus: PENDING -> APPROVED
Activity diagram (Report 4 section)	Report 4 Section 3.2.2 (Activity Diagram AD-WF02)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-05	Category-Locked 6-Block Product & Compliance Moderation	Product category locks to store.category (mismatch returns 400); duplicate EAN-13 barcode fails; new SKU stays DRAFT until a System Manager approves (APPROVED).	TC-F05-01..03
F-06	MAP Floor-Price Guard & Price Promotion Approval	System Manager sets SKU floorPrice; single/batch promotion with proposedPrice < floorPrice returns 400; discount applies only when status=ACTIVE within budget.	TC-F06-01..03
F-07	4-Axis AI Partner Matchmaking & Two-Way Cooperation	AI scores verified KOCs (0–100 across Category, CR%, Tier/Social, Price Fit) in <=500ms; bilateral partnership invite/request transitions PENDING -> APPROVED.	TC-F07-01..03
F-08	Centralized Digital Media Asset Hub	Shop Manager uploads product-linked HD images/videos/copy; verified Collaborator filters by type and downloads in 1 click; unverified access returns 403.	TC-F08-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02a	**Main**	Sora Skincare (locked to Cosmetics), verified KOC profiles, and SKU floorPrice = 350,000 VND.	1. Shop Manager creates Vitamin C Serum (450k VND, EAN-13); System Manager approves. 2. Shop Manager proposes 390k VND promotion (>= floorPrice); System Manager approves. 3. Run 4-Axis AI Matchmaker, approve KOC cooperation, and KOC downloads HD media.	Product APPROVED on storefront; promotion ACTIVE at 390k VND; AI returns ranked KOC radar scores; StoreCollaborator=APPROVED and HD asset downloaded.	Phan Xuân Thịnh
DS-02b	**Exception**	Shop Manager attempts cross-category SKU, duplicate EAN-13 barcode, or price drop below floorPrice (320k < 350k VND).	1. Submit Electronics category from Cosmetics store. 2. Submit duplicate EAN-13 barcode. 3. Propose promotion price 320,000 VND below 350,000 VND MAP floor.	All three requests rejected with 400 Bad Request; product and price remain unchanged, protecting brand vertical isolation and unit margin.	Phan Xuân Thịnh

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	16	Yes (>= 90%)	Phan Xuân Thịnh
End-to-end workflow test (>= 1)	1	Yes (100%)	Phan Xuân Thịnh
State transition tests (>= 1 per transition)	3	Yes (100%)	Phan Xuân Thịnh
NFR & security measurement tests (>= 1 per NFR)	2	Yes (100%)	Phan Xuân Thịnh
Total for this workflow	**22**	**Yes (>= 90%)**	**Phan Xuân Thịnh**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.3 Workflow WF-03

Workflow name	WF-03: 14-Day Sample Experience SLA, Multi-Channel Marketing Tools (Smart Link, Dynamic QR, Exclusive Coupon) & Authenticated Checkout
Owner (one member)	Nguyễn Đình Tuấn
Supporting members	Phan Xuân Thịnh, Nguyễn Thành Thắng
Actors	Shop Manager, Collaborator, System Administrator, Customer
Trigger	Collaborator requests 0 VND product sample (with 14-day review video SLA), generates Smart Link, Dynamic QR, and Exclusive KOL Coupon (Web & Mobile); Customer places authenticated order.
Business outcome	Sample dispatched with 14-day SLA tracking; anti-self-referral guard blocks creator self-commission; multi-channel attribution captured with zero loss.
Main entity state path	ExclusiveDealStatus: PENDING -> PENDING_ADMIN_APPROVAL -> APPROVED; SampleRequestStatus: PENDING -> APPROVED -> SHIPPED -> RECEIVED -> VIDEO_SUBMITTED -> COMPLETED / OVERDUE; LiveSessionStatus: SCHEDULED -> LIVE -> ENDED
Activity diagram (Report 4 section)	Report 4 Section 3.2.3 (Activity Diagram AD-WF03)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-09	Socket.IO Realtime VIP Deal Negotiation (10%–16%)	Counterparties send/respond to rich VIP Deal cards (targetRevenue, commitmentDays) over /chat WebSocket without reload; 10%–16% deals activate immediately on accept.	TC-F09-01..03
F-10	VIP Budget Exception Escalation Gate (17%–25%)	Proposals >25% fail with 400; accepted 17%–25% deals transition to PENDING_ADMIN_APPROVAL and activate only after SYSTEM_ADMIN approves at /admin/vip-exceptions.	TC-F10-01..03
F-11	Sample Product Sponsorship & 14-Day Video Review SLA	Verified KOC requests 0 VND sample; GHN receipt (RECEIVED) starts 14-day SLA timer; unsubmitted video past deadline marks OVERDUE and blocks new sample requests.	TC-F11-01..03
F-12	Value-Added Livestream Room & Live Deal Pinning	Creator transitions SCHEDULED -> LIVE -> ENDED and pins store SKUs (+2% live bonus); stock=0 disables buy CTA in live tray; ENDED locks pinning and shows summary.	TC-F12-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-03a	**Main**	Approved Sora–KOC partnership on Web/Mobile; approved Vitamin C Serum with sample quota > 0.	1. Shop sends 16% VIP Deal card (KPI 30M VND) via Socket.IO Chat; KOC accepts (auto-APPROVED); also test 18% deal escalating to Admin approval. 2. KOC requests sample -> Shop dispatches GHN -> KOC submits review URL within 14d -> Shop marks COMPLETED. 3. KOC pins SKU in LIVE room.	16% VIP deal active immediately (18% active after Admin approval); sample reaches COMPLETED within 14-day SLA; LIVE tray pins SKU with +2% bonus.	Nguyễn Đình Tuấn
DS-03b	**Exception**	VIP Deal proposed at 28% (>25% cap); KOC misses 14-day sample review deadline; pinned SKU stock hits 0.	1. Propose 28% VIP Deal in chat. 2. Trigger overdue scan on sample past 14-day deadline and request another sample. 3. Set pinned SKU stock = 0 during LIVE session.	28% deal blocked (400 Bad Request); sample transitions to OVERDUE and blocks new sample requests; out-of-stock SKU disables purchase CTA in live tray.	Nguyễn Đình Tuấn

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	15	Yes (>= 90%)	Nguyễn Đình Tuấn
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Đình Tuấn
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Đình Tuấn
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Đình Tuấn
Total for this workflow	**20**	**Yes (>= 90%)**	**Nguyễn Đình Tuấn**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.4 Workflow WF-04

Workflow name	WF-04: Multi-Warehouse GHN Logistics Sync, 14-Day Dual-Path Returns (Exchange vs Refund Clawback) & Deal KPI Tracking
Owner (one member)	Nguyễn Phú Quý

Supporting members	Nguyễn Đình Tuấn, Phan Xuân Thịnh
Actors	Collaborator, Customer, Shop Manager, System Manager
Trigger	Collaborator shares Smart Link/QR and KOC Attribution Coupon; Customer checks out, receives GHN parcel, and files either a damaged-item Exchange or a Refund return.
Business outcome	Order fulfilled via dedicated brand GHN warehouse; delivery webhook triggers 14-day return window and hold; 1-for-1 EXCHANGE preserves 100% commission (+ carrier compensation claim) while REFUND claws back commission to 0 VND and deducts KPI GMV.
Main entity state path	OrderStatus: PENDING -> SHIPPING -> DELIVERED -> COMPLETED / RETURNED; ReturnRequestStatus: REQUESTED -> SHOP_APPROVED -> INSPECTING -> EXCHANGE_SHIPPED / REFUNDED -> COMPLETED; CommissionStatus: PENDING -> APPROVED / REVERSED
Activity diagram (Report 4 section)	Report 4 Section 3.2.4 (Activity Diagram AD-WF04)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-13	Smart Link, Dynamic QR & KOC Attribution Coupon	Verified KOC creates /r/:shortCode link, QR & goods coupon ([KOC][5|10], shipping vouchers excluded); click sets 30d AttributionSession rate-limited at 100 req/min/IP.	TC-F13-01..03
F-14	Authenticated Zero-Loss Checkout & Self-Referral Guard	Checkout requires login while preserving 100% cart items and settles via PayOS VietQR/Wallet/COD; if buyer matches KOC (isSelfReferral), commission is forced to 0%.	TC-F14-01..03
F-15	Multi-Warehouse GHN Fulfillment & Order Tracking	Shop Manager confirms order and dispatches GHN parcel from brand warehouse (weight/dimensions); SKU stock decrements atomically and status tracks SHIPPING -> DELIVERED.	TC-F15-01..03
F-16	Dual-Path Return Resolution: Exchange vs Refund Clawback	Within 14-day return window & 48h inspection SLA: EXCHANGE ships 1-1 replacement, preserves 100% KOC commission & opens GHN claim; REFUND credits wallet & reverses commission.	TC-F16-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-04a	**Main**	Verified KOC with active 16% VIP deal & KOC coupon (10% off); authenticated Customer; Sora GHN warehouse.	1. KOC generates /r/:shortCode, QR & KOC coupon. 2. Customer clicks link / applies KOC coupon, logs in preserving 100% cart, pays via PayOS/COD. 3. Shop creates GHN waybill -> DELIVERED. 4. On transit damage, Customer chooses EXCHANGE -> Shop ships 1-1 replacement & opens GHN claim.	Order attributed to KOC with pending commission recorded; replacement delivered (COMPLETED), 100% KOC commission preserved, and GHN compensation claim opened.	Nguyễn Phú Quý
DS-04b	**Exception**	KOC attempts self-purchase via own link/coupon; Customer requests full return & refund within 14-day window.	1. KOC checks out using own referral link/matching phone. 2. Separate Customer files REFUND return with unboxing video -> parcel returned & inspected -> refund approved.	Self-referral order sets commission rate = 0% and flags AiFraudService; refunded order credits Customer shoppingBalance and reverses KOC commission to REVERSED.	Nguyễn Phú Quý

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	15	Yes (>= 90%)	Nguyễn Phú Quý
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Phú Quý
State transition tests (>= 1 per transition)	3	Yes (100%)	Nguyễn Phú Quý
NFR & security measurement tests (>= 1 per NFR)	1	Yes (100%)	Nguyễn Phú Quý
Total for this workflow	**20**	**Yes (>= 90%)**	**Nguyễn Phú Quý**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

6.5 Workflow WF-05

Workflow name	WF-05: Two-Stage Commission Settlement (14-Day Hold & KPI Escrow), Row-Locked Payouts with 10% PIT & Executive Ledger
Owner (one member)	Nguyễn Thành Thắng
Supporting members	Nguyễn Đình Tuấn, Nguyễn Phú Quý
Actors	Shop Manager, Collaborator, System Manager, System Administrator
Trigger	14-day return window elapses on delivered orders, closed-month milestone settlement evaluates, active Exclusive Deal achieves 100% KPI, and Collaborator requests bank withdrawal.
Business outcome	Stage 1 (10% + Tier%) unlocks after 14 days clean delivery; Stage 2 KPI differential unlocks upon 100% KPI achievement; 10% flat PIT withheld under DB row lock; net payout disbursed via Napas 24/7; audited in Executive Financial Ledger.
Main entity state path	CommissionStatus: PENDING -> APPROVED; DealBonusStatus: PENDING_KPI -> QUALIFIED / FORFEITED; PayoutStatus: PENDING -> PROCESSING -> APPROVED / REJECTED
Activity diagram (Report 4 section)	Report 4 Section 3.2.5 (Activity Diagram AD-WF05)

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F-17	Best-Benefit Commission, 14-Day Hold & KPI Deal Escrow	Computes min(25%, max(VIP Deal, 10%+Tier, 10%) + Live); holds base fee 14 days in pendingBalance and holds VIP bonus in conditionalDealEscrowBalance until KPI QUALIFIED.	TC-F17-01..03
F-18	Monthly Milestone Settlement (FR-09) & Tier Promotion	Closed-month clean GMV scan excludes refunded/self-referral orders, pays 200k–4M VND milestone bonus idempotently (0 duplicates), and promotes Bronze->Silver->Gold->Diamond.	TC-F18-01..03
F-19	Row-Locked Collaborator Withdrawal, 10% PIT & Napas 24/7 Payout	SHOP_MANAGER withdrawal returns 403; COLLABORATOR request (>=200,000 VND) locks wallet row, withholds flat 10% PIT (FinancePosting), and disburses net via Napas 24/7 Sandbox.	TC-F19-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05a	**Main**	Delivered order past 14-day hold with no open return; closed-month clean GMV > 50M VND; verified KOC wallet.	1. Reconciliation job matures 14-day PENDING commission to availableBalance and qualifies KPI deal escrow. 2. Shop settles closed-month FR-09 milestone bonus & promotes KOC tier. 3. KOC requests 2,072,000 VND withdrawal -> System Manager approves via Napas 24/7 Sandbox.	Wallet debited once under SELECT ... FOR UPDATE; 207,200 VND (10% PIT) posted via FinancePosting; 1,864,800 VND net approved with SBX-... reference and AuditLog.	Nguyễn Thành Thắng
DS-05b	**Exception**	SHOP_MANAGER attempts withdrawal; KOC requests < 200,000 VND or fires parallel requests exceeding balance; Napas 24/7 Sandbox timeout.	1. Call POST /payouts/withdraw as SHOP_MANAGER. 2. Submit 100,000 VND (<200k min) and 2 concurrent full-balance requests. 3. Simulate BIDV timeout/rejection.	SHOP_MANAGER blocked (403 Forbidden); <200k rejected (400); row lock allows at most 1 request (0 negative balance); rejected payout restores 100% funds via PAYOUT_REJECT_REFUND.	Nguyễn Thành Thắng

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests (>= 3 per feature)	15	Yes (>= 90%)	Nguyễn Thành Thắng
End-to-end workflow test (>= 1)	1	Yes (100%)	Nguyễn Thành Thắng

Test type	Count	Automated in CI	Responsible member(s)
State transition tests (>= 1 per transition)	2	Yes (100%)	Nguyễn Thành Thắng
NFR & security measurement tests (>= 1 per NFR)	2	Yes (100%)	Nguyễn Thành Thắng
Total for this workflow	**20**	**Yes (>= 90%)**	**Nguyễn Thành Thắng**

Pass criteria at Defense 1.1

Runs end-to-end on staging with seeded data	Yes (Web Portal & Android Mobile API)
Main scenario passes	Yes (100%)
Exception scenarios passing	100% (1/1)
Test pass rate for this workflow	>= 95% (Target >= 90%)
Open Critical / High defects	0

Reviewer note

Reviewer note	

7. Individual Commitments

7.1 Member 1

Full name	Nguyễn Thành Thắng
Student ID	SE184251
Role in team	Team Leader, Fullstack Architect & Brand Governance Lead

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-01: Internal Brand Staff Provisioning, KOC eKYC Onboarding & Social Verification; WF-05: Two-Stage Commission Settlement (14-Day Hold & KPI Escrow), Row-Locked Payouts with 10% PIT & Executive Ledger	Sole workflow owner
Mandatory features owned	F-01 (Auth/Email-OTP/RBAC), F-02 (Internal Staff & Stores), F-03 (KOC eKYC & Social), F-04 (AuditLog), F-17 (Two-Stage Commission & 14-Day Hold), F-18 (Monthly Milestone Settlement FR-09), F-19 (Row-Locked Payout & 10% PIT)	Disjoint feature ownership
Modules / entities owned	AuthModule, UsersModule, StoresModule, KycModule, AuditModule, CommissionsModule, WalletsModule, PayoutsModule	Code & unit/E2E tests
Design decisions owned	D1: In-House Brand Provisioning & Shared Mobile/Web Bootstrap; D2: MAP Floor-Price Guard & KOC Attribution Coupon	BM01 Section 3
NFRs responsible	NFR-01 (API & Redirect Latency <= 200ms), NFR-02 (Argon2id & 5-Role RBAC Isolation)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	38 committed functional test cases (WF-01: 18 + WF-05: 20)	Defense 1.1
Of which automated in CI	38 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-01a, DS-01b, DS-05a, DS-05b	Defense 1.1
Required fixes owned	G1 (Sync SDD 69-model ERD & System Overview), G4 (Clean legacy deprecated stubs)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-01, WF-05	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ **Verified** ☐ **Partly verified** ☐ **Not verified**		

Member confirmation

Signature / date	Confirmed by Nguyễn Thành Thắng (SE184251) · 06/11/2026

7.2 Member 2

Full name	Phan Xuân Thịnh
Student ID	SE184527
Role in team	Fullstack Developer, Realtime & Mobile Lead

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-02: Category-Locked Product Compliance, MAP Floor-Price Promotion, AI Matchmaking & Socket.IO Deal Negotiation	Sole workflow owner

Type	Items (IDs / names)	Notes
Mandatory features owned	F-05 (Category-Locked Catalog & Moderation), F-06 (MAP Floor-Price Guard), F-07 (4-Axis AI Matchmaking), F-08 (Socket.IO Deal Chat & Realtime Media/File Sharing)	Disjoint feature ownership
Modules / entities owned	ProductsModule, PricingModule, AiModule, ChatModule	Code & unit/E2E tests
Design decisions owned	D2: MAP Floor-Price Guard & Category Lock; D3: Bi-Directional Exclusive Deal Negotiation via Socket.IO	BM01 Section 3
NFRs responsible	NFR-03 (Web Bundle <= 500KB / FCP <= 1.5s & Android Mobile Bootstrap <= 300ms)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	22 committed functional test cases	Defense 1.1
Of which automated in CI	22 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-02a, DS-02b	Defense 1.1
Required fixes owned	G2 (Mobile native checkout handoff & staging Android APK verification)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-02	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ Verified ☐ Partly verified ☐ Not verified		

Member confirmation

Signature / date	Confirmed by Phan Xuân Thịnh (SE184527) · 06/11/2026

7.3 Member 3

Full name	Nguyễn Phú Quý
Student ID	SE180104
Role in team	Fullstack Developer, Attribution & Reverse Logistics Lead

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-04: Multi-Warehouse GHN Logistics Sync, 14-Day Dual-Path Returns (Exchange vs Refund Clawback) & Deal KPI Tracking	Sole workflow owner
Mandatory features owned	F-15 (GHN Logistics Status Ingestion & 14-Day Hold Sync), F-16 (Post-Sale Remuneration: Exchange Protection vs Refund Clawback)	Disjoint feature ownership
Modules / entities owned	ShippingModule (GHN Express), ReturnsModule, OrdersModule	Code & unit/E2E tests
Design decisions owned	D4: 14-Day Return Window & Dual-Path Return Resolution (1-1 Exchange vs Refund Clawback)	BM01 Section 3
NFRs responsible	NFR-04 (100% Atomic Checkout & Return State Transition Rollback)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	20 committed functional test cases	Defense 1.1
Of which automated in CI	20 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-04a, DS-04b	Defense 1.1
Required fixes owned	0 open High defects (GHN compensation claim & dual-path return integration verified)	Before Defense 1.1

Commitment	Committed (count / IDs)	Due
Documentation sections	SRS/SDD sections for WF-04	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ **Verified** ☐ **Partly verified** ☐ **Not verified**		

Member confirmation

Signature / date	Confirmed by Nguyễn Phú Quý (SE180104) · 06/11/2026

7.4 Member 4

Full name	Nguyễn Đình Tuấn
Student ID	SE182540
Role in team	Backend Lead & Financial Settlement Specialist

Ownership

Type	Items (IDs / names)	Notes
Main workflow(s) owned	WF-03: 14-Day Sample Experience SLA, Multi-Channel Marketing Tools (Smart Link, Dynamic QR, Exclusive Coupon) & Authenticated Checkout	Sole workflow owner
Mandatory features owned	F-09 (Socket.IO Deal Negotiation <=16%), F-10 (>16% Admin Gate), F-11 (Sample Sponsorship & 14-Day Video SLA), F-12 (Peak-Hour Marketing Sessions), F-13 (Smart Link, Dynamic QR & Exclusive Coupon), F-14 (Anti-Self-Referral Checkout Guard)	Disjoint feature ownership
Modules / entities owned	SamplesModule, ReferralLinksModule, CouponsModule, CheckoutModule, MobileModule & Expo Android App	Code & unit/E2E tests
Design decisions owned	D1: Mobile Companion App & Smart Attribution Tools; D3: 14-Day Sample SLA Enforcement	BM01 Section 3
NFRs responsible	NFR-05 (100 req/min/IP Rate Limit & Redis Fallback), NFR-06 (Row-Lock Concurrency & 100% Ledger Auditability)	BM01 Section 7

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Functional test cases written	20 committed functional test cases	Defense 1.1
Of which automated in CI	20 (>= 90% automated in GitHub Actions CI)	Defense 1.1
Demo scenarios presented	DS-03a, DS-03b	Defense 1.1
Required fixes owned	G3 (Maintain Napas 24/7 Sandbox idempotency/inquiry adapter & FinancePosting ledger docs)	Before Defense 1.1
Documentation sections	SRS/SDD sections for WF-03	Before Defense 1.1

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
☐ **Verified** ☐ **Partly verified** ☐ **Not verified**		

Member confirmation

Signature / date	Confirmed by Nguyễn Đình Tuấn (SE182540) · 06/11/2026

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF-01	**Nguyễn Thành Thắng**	F-01, F-02, F-03, F-04	DS-01a (Main), DS-01b (Exception)	18	Yes (>= 90%)
WF-02	**Phan Xuân Thịnh**	F-05, F-06, F-07, F-08	DS-02a (Main), DS-02b (Exception)	22	Yes (>= 90%)
WF-03	**Nguyễn Đình Tuấn**	F-09, F-10, F-11, F-12, F-13, F-14	DS-03a (Main), DS-03b (Exception)	20	Yes (>= 90%)
WF-04	**Nguyễn Phú Quý**	F-15, F-16	DS-04a (Main), DS-04b (Exception)	20	Yes (>= 90%)
WF-05	**Nguyễn Thành Thắng**	F-17, F-18, F-19	DS-05a (Main), DS-05b (Exception)	20	Yes (>= 90%)
Total	**4 members (1 sole owner/WF)**	**18 features (F-01..F-19)**	**10 scenarios (5M / 5E)**	**100**	**>= 90%**

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
Nguyễn Thành Thắng (Leader)	WF-01, WF-05	F-01..F-04, F-17..F-19	38	Yes (>= 90%)	DS-01a, DS-01b, DS-05a, DS-05b	☐ Verified
Phan Xuân Thịnh	WF-02	F-05, F-06, F-07, F-08	22	Yes (>= 90%)	DS-02a, DS-02b	☐ Verified
Nguyễn Phú Quý	WF-04	F-15, F-16	20	Yes (>= 90%)	DS-04a, DS-04b	☐ Verified
Nguyễn Đình Tuấn	WF-03	F-09, F-10, F-11, F-12, F-13, F-14	20	Yes (>= 90%)	DS-03a, DS-03b	☐ Verified
Total	**5 workflows (WF-01..05)**	**18 features (F-01..F-19)**	**100**	**>= 90%**	**10 scenarios**	**☐ Verified**

8.3 Committed Numbers

Number of main workflows	5 (WF-01..WF-05; 1 sole owner per workflow)
Number of demo scenarios (main / exception)	10 (5 Main / 5 Exception)
Committed functional test cases	81 (57 Feature + 5 E2E + 13 State + 6 NFR; 0 double-counted)
Percentage automated in CI	>= 90% automated in GitHub Actions CI (Minimum requirement >= 50%)

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	100% run end-to-end	100% (5/5 workflows run end-to-end on Cloud Staging & Mobile API)
Demo scenarios	100% main; >= 80% exception	100% main (5/5) and 100% exception (5/5) reproducible via seed script
Committed test cases pass rate	>= 90% at Defense 1.1	>= 95% overall pass rate; 100% pass rate on mandatory main-flow tests
Automated in CI	>= 50% of committed total	>= 90% of committed 81 test cases executed automatically in CI
Open Critical / High defects	0 open Critical or High defects	0 open Critical or High defects at Defense 1.1
SRS/SDD & Self-Review alignment	No unexplained High findings	All BM02 Planned items (SR-01..SR-04) closed with commit links

Criterion	Minimum threshold	Agreed threshold
Individual ownership	Every member demos owned workflow	Each of the 4 members demos and defends their owned workflow(s) live

9. Required Fixes

ID	Item	Related workflow	Owner	Due
FX-01 (SR-01)	Regenerate Report 4 Logical ERD & Data Dictionary to match all 69 Prisma models and 5 UserRole values	WF-01, WF-02	Nguyễn Thành Thắng	20/11/2026
FX-02 (SR-02)	Update Report 4 System Overview Diagram & Tech Table (React/Vite, Expo Android, @nestjs/schedule, BIDV/PayOS/GHN)	WF-01	Nguyễn Thành Thắng	20/11/2026
FX-03 (SR-03)	Hide legacy Mobile menu stub and mark deprecated database table in SDD after scope cleanup	WF-02	Nguyễn Thành Thắng	20/11/2026
FX-04 (SR-04)	Verify Web-checkout handoff, staging Cloud deployment, and 5-role Android APK E2E test run	WF-03	Phan Xuân Thịnh	25/11/2026

10. Review 2 Conclusion (completed by reviewers)

☐ Pass ☐ Conditional pass ☐ Re-review

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1	**longt5**		06/11/2026
Reviewer 2	**ThinhDP2**		06/11/2026
Supervisor	**ThS. Tôn Thất Hoàng Minh**		06/11/2026
Team leader	**Nguyễn Thành Thắng**		06/11/2026